

MoveInSync Hackathon Problem Statement

1. Problem Statement Title

Agentic Intelligence & Reporting Layer for Enterprise Mobility

2. Theme

Agentic AI (domain: Enterprise Mobility / Operations Intelligence)

3. Background / Business Context

Large enterprises move a few hundred to a few thousand employees daily through a mix of home and nodal pick-and-drop cabs and fixed-route shuttles. Transport managers are accountable for cost, safety, experience, and sustainability — but most of their time goes into assembling data, not acting on it. The signal is rich. The insight is missing. The actions are manual.

Three personas feel this differently:

- **Transport manager** (operational) — owns day-to-day ops: vendor coordination, escalations, shift planning, delay management. Needs fast, actionable signals, not reports.
- **Transport & facilities head** (strategic) — owns budget, SLA accountability, vendor strategy, leadership reporting. Needs a coherent cost/safety/experience story without assembling it manually.
- **Team / line manager** (shift-based ops) — needs shift-level visibility into who made it, who was late, and how delays ripple into floor/ops readiness.

4. Problem Definition

Mobility operations generate rich structured data continuously (trips, vendors, drivers, employees — ops scale, timeliness/delays, safety/compliance, cost, sustainability, employee experience, vendor performance). Today this largely sits in static weekly/monthly reports.

Pain point: a metric without context is just a number. E.g. "OTA is 78%" matters far less than "it was 85% last month, SLA is 90%, and two vendors are responsible for the gap." Benchmarking against historical trend / SLA / industry norm / peer is currently absent.

Desired improvement: an agentic layer that senses what's happening, reasons about what it means, and acts — with minimal human prompting — surfacing what matters, benchmarking it, and communicating outcomes to the right person at the right time.

5. Tech Stack

Open / participant's choice — *preferably Java, Angular, AWS resources, but not restrictive.*

6. Resources Provided

Sample dataset only: anonymised trip logs across cab, nodal, and shuttle modes — including vendor performance, GPS traces, delay records, cost data, and employee feedback.

7. Expected Output

Solutions may take any of the following forms, or combine several:

- Conversational agent (NL Q&A on mobility data)
- Proactive alerting & triggers
- Automated reporting & narratives
- Insight & anomaly detection
- Decision-support dashboard
- Automated communications

8. Requirements

Mandatory

- Working, demo-able prototype running on the provided dataset
- Agentic behaviour — senses, reasons, and acts; not a passive dashboard or query-only tool
- Serves at least one of the three named personas
- Contextualises metrics against at least one reference point (historical trend, SLA/goal, industry benchmark, or peer comparison)

Good-to-have

- Combines two or more of the solution forms listed in Section 7
- Handles messy or missing data gracefully (GPS gaps, unmatched records, incomplete rosters)
- Proactive triggers rather than purely on-demand responses

Bonus

- Credible deployability story into an existing enterprise mobility platform — multi-tenancy, latency, cost
- Output a transport & facilities head could forward to leadership without rework

Not expected

- Production-grade authentication or security
- A full historical data pipeline
- Integration with real vendor systems

Constraints

- Anonymised sample trip-log dataset only — no live system access

9. Evaluation Criteria

#	Criteria	What we are looking for	Weight
1	Business impact & experience	Does it meaningfully reduce manager effort or surface decisions that would otherwise be missed — and does it land? Clarity for the intended persona, leadership-ready output, shareable without rework.	35
2	Agentic design & cost at scale	Is AI solving a genuine problem rather than decorating the solution — and can it actually be run? Inference cost per interaction, latency, efficiency at enterprise volumes.	20

#	Criteria	What we are looking for	Weight
3	Architecture & code quality	Sound structure and engineering judgement; deployable into an existing platform; choices a team could build on.	20
4	Functionality	It runs — a working, demo-able prototype, end to end, on the provided dataset.	25
	Total		100

10. Deliverables

- Source code repository (GitHub/GitLab)
- Architecture diagram
- README + setup instructions
- Sample inputs/outputs
- Demo video (if requested)
- Presentation deck
- Live demo

Requirements Mandatory

- It runs on the sample dataset.
- It senses, reasons, and acts.
- It serves one of the three personas.
- And every metric carries context.

Good-to-have

- It handles messy data.
- It triggers on its own.
- It combines two or more solutions mentioned earlier.

Bonus

- An output a facilities head could forward to leadership, untouched.



Conversational agent
(NL Q&A on mobility data)

Proactive alerting & triggers

Automated reporting & narratives

Insight & anomaly detection

Decision-support dashboard

Automated communications



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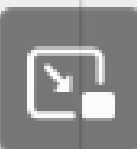
Selection Criteria

- Business impact (35 Points)
- Functionality (25 Points)
- Agentic design & cost at scale (20 Points)
- Architecture & code quality (20 Points)

Deliverables



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also keep in mind that they are grading in terms of

1. whether it works
2. whether it actually lands
3. whether it can actually be actually run on an enterprise scale

tech stack they mentioned is Java angular AWS